

Prompt Evaluation & Hallucination Mitigation

A technical deep-dive into measuring AI output quality and building reliable LLM systems – from evaluation metrics to hallucination reduction strategies.

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What is Prompt Evaluation?

A **prompt** is the instruction given to an AI model – e.g., *"Explain machine learning in simple English for a first-year engineering student."* **Prompt evaluation** is the systematic process of measuring how well a prompt produces the required answer.

Correctness

Is the information factually accurate?

Relevance

Does the answer address the question?

Completeness

Are all required parts covered?

Instruction Following

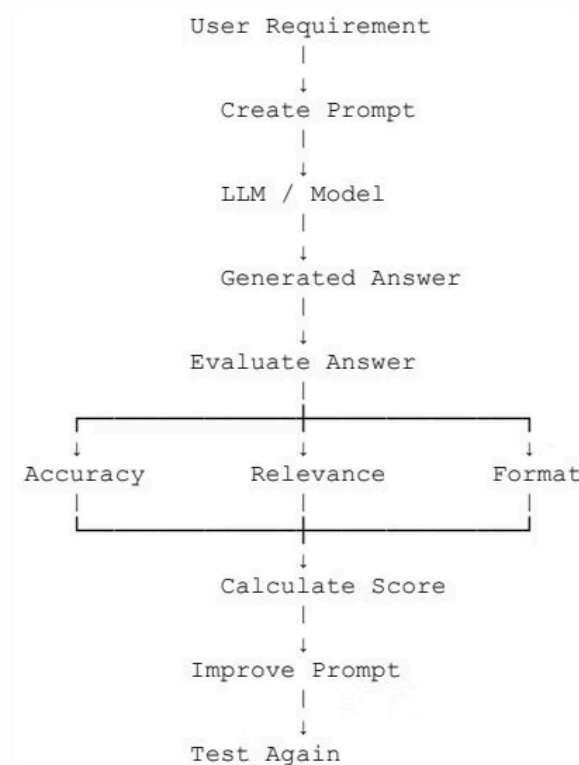
Did the model obey all constraints?

Safety

Is the output free from harmful content?

The Prompt Evaluation Flow

Prompt evaluation is an **iterative process**. A prompt is created, sent to the LLM, and the generated answer is evaluated across multiple dimensions. Scores drive prompt improvement, and the cycle repeats until quality targets are met.



Why Iterate?

A vague prompt like *"Tell me about Python"* produces a general answer. A structured prompt with scope, format, and audience constraints produces a measurably better result — and evaluation proves it.

Prompt A vs Prompt B

Metric	Prompt A	Prompt B
Relevance	3.8	4.8
Completeness	3.2	4.9
Overall	3.75	4.83

Evaluation Criteria & Automatic Methods

Key Evaluation Criteria

Correctness

Factually accurate output
(e.g., capital of India = New Delhi ✓)

Fluency

Grammatically correct,
readable prose

Consistency

Same prompt → similar
quality answers across runs

Safety

No harmful, biased, or private
information

Automatic Evaluation Methods

1 Exact Match

Score = 1 if output matches expected exactly; 0 otherwise. Fails for paraphrased correct answers.

2 Semantic Similarity

Embedding model converts sentences to vectors; cosine similarity measures meaning overlap regardless of wording.

3 LLM-as-a-Judge

A second LLM evaluates the first LLM's answer on correctness, relevance, and completeness, returning a JSON score.

📌 Weighted Score Formula: **Overall = 0.4×Correctness + 0.3×Relevance + 0.2×Completeness + 0.1×Clarity**

AI Hallucination — What & Why

An **AI hallucination** occurs when a model generates information that appears believable but is actually **false, fabricated, or unsupported**. LLMs do not query a verified database — they predict the next token based on learned patterns, so they can confidently produce incorrect facts.

Factual

"The Earth has two moons." — incorrect facts stated confidently.

Fabricated References

Invented papers, authors, URLs, or statistics that do not exist.

Numerical

Wrong numbers that look authoritative — e.g., "India's population is 2.7 billion."

Contextual

Model ignores provided context and introduces unrelated information.

Instruction

Model claims to have checked a database when it did not.

Hallucination Mitigation — Core Techniques



Better Prompting

Instruct the model: *"Answer ONLY using the information below. If insufficient, say 'I don't have enough information.' Do not invent facts."*
This establishes a grounding rule.



Grounding & RAG

Provide trusted context before the question. RAG (Retrieval-Augmented Generation) retrieves relevant documents from a vector database and supplies them as context to the LLM, reducing reliance on internal knowledge.



Tool Calling

Route calculations, lookups, and API calls to external tools (calculators, databases, search engines) instead of expecting the LLM to compute them internally — far more reliable.



Structured Output & Validation

Request JSON with required fields to constrain responses. After generation, a verifier checks whether claims are supported; unsupported answers are regenerated or rejected.

RAG Architecture & Its Limits

RAG Pipeline



Example: A college chatbot retrieves the attendance rule from a PDF and answers: *"Minimum attendance is 75%. Source: Academic Regulations 2026, Page 14."*

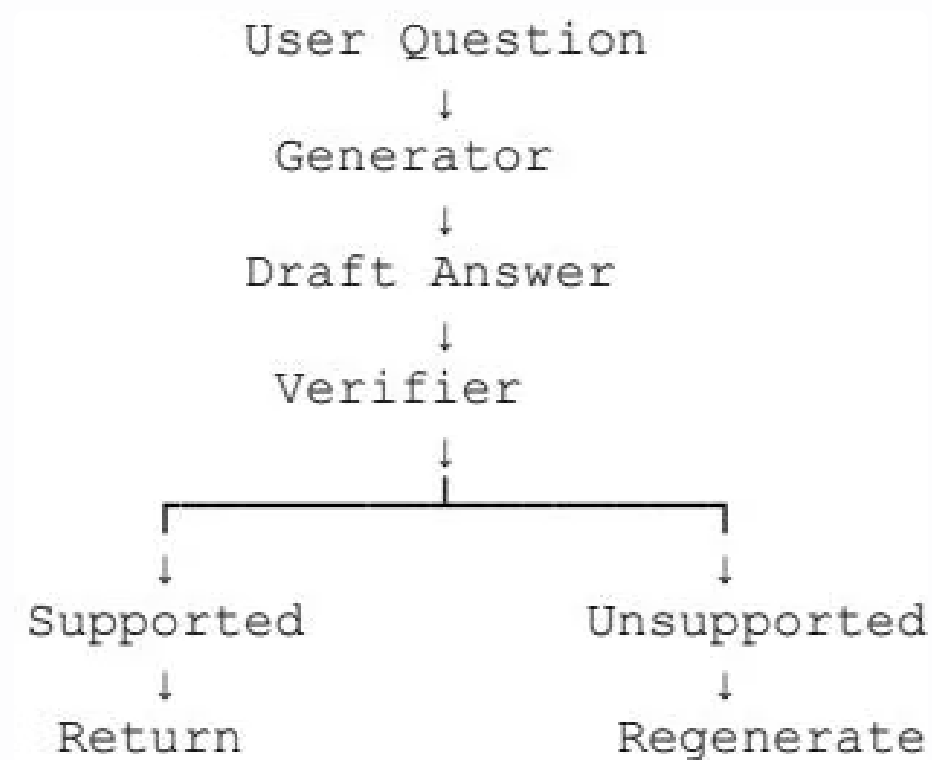
RAG Does NOT Guarantee Zero Hallucination

- Wrong Document Retrieved
Retrieval may return an irrelevant chunk.
- Incomplete Context
Retrieved text may be truncated or partial.
- LLM Misinterpretation
Model may misread the supplied context.
- Residual Generation
LLM may still add unsupported information.

❏ **Formula for reliability:** Good Retrieval + Good Prompting + Verification = More Reliable System

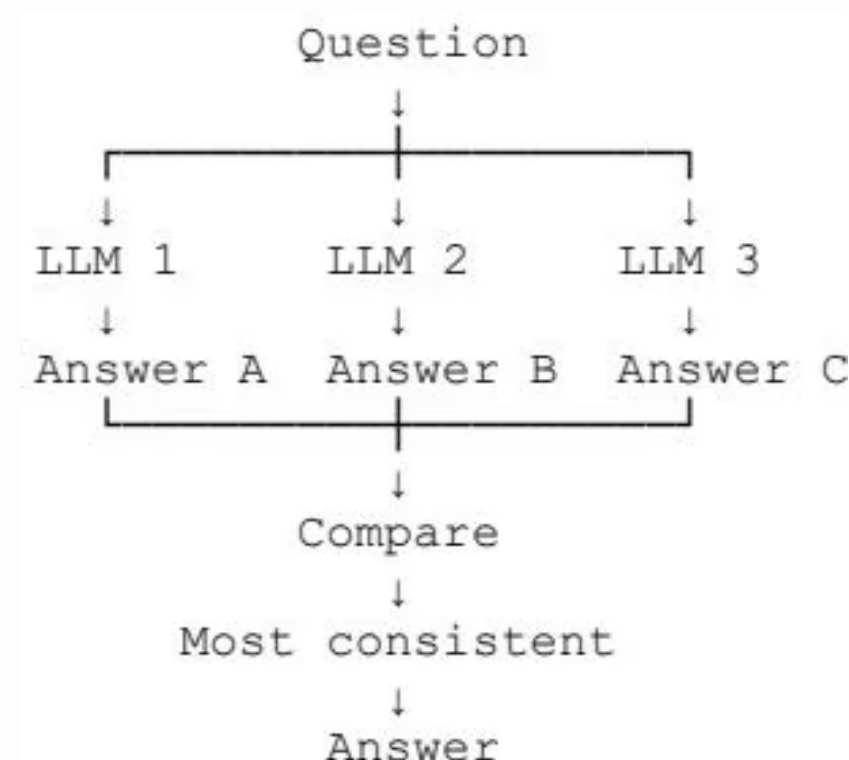
Advanced Architectures for Reliability

Two-Stage LLM (Generator + Verifier)



Generator produces a draft; Verifier checks if claims are supported. Unsupported answers are regenerated – not returned to the user.

Self-Consistency



The same question is sent to multiple LLM instances. Answers are compared and the most consistent response is selected – improving reliability for complex reasoning tasks.

System A — Answers Everything

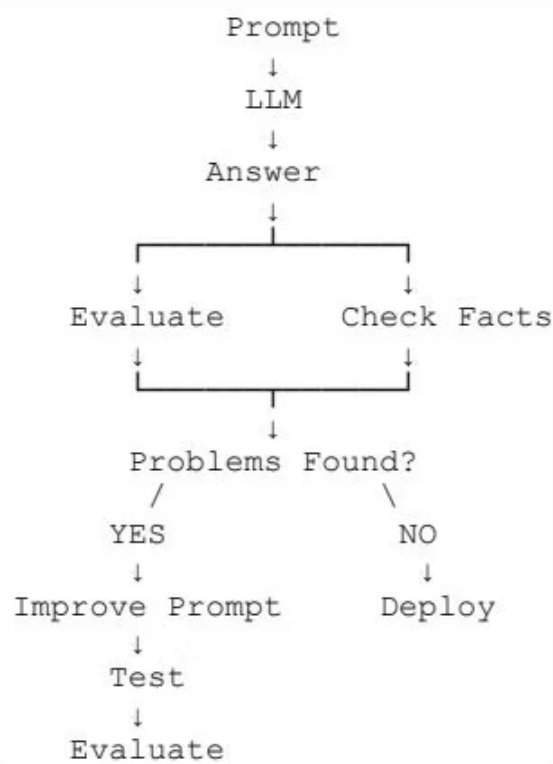
Accuracy: **70%** — dangerous in high-stakes domains

System B — Answers Only When Evidence Exists

Accuracy: **95%** — "I don't know" is a feature, not a failure

Prompt Evaluation + Hallucination Mitigation Together

The Combined Improvement Cycle



Evaluation measures output quality; hallucination checks verify factual grounding. Together they drive continuous prompt improvement before deployment.

Memory Mnemonics

C-R-C-I-C-S —
Prompt Evaluation

Correctness · **R**elevance ·
Completeness ·
Instruction Following ·
Clarity · **S**afety

P-G-R-T-V-H —
Hallucination
Mitigation

Prompt Engineering ·
Grounding · **RAG** · **T**ools ·
Verification · **H**uman
Evaluation

Key Distinction

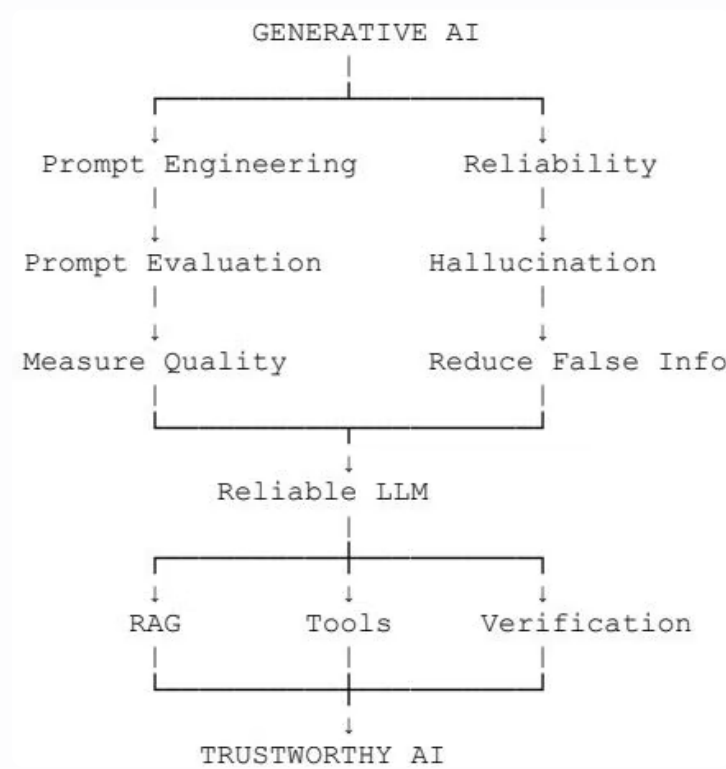
Prompt Evaluation

"How good is my
prompt/output?"

Hallucination
Mitigation

"How do I reduce false
information?"

From Generative AI to Trustworthy AI



Exam-Oriented Summary

01

Prompt Engineering

Craft specific, grounded, structured prompts with clear constraints and audience context.

02

Prompt Evaluation

Measure quality using human scoring, exact match, semantic similarity, BLEU/ROUGE, and LLM-as-a-Judge.

03

Hallucination Mitigation

Apply RAG, grounding, tool calling, structured output, source citation, and self-consistency.

04

Continuous Evaluation

Build a dataset of test Q&A pairs, score every prompt version, and iterate until deployment-ready.

❏ Trustworthy AI = Reliable LLM + RAG + Tools + Verification + Continuous Human Oversight